

Saddam Khan

Machine Learning Researcher

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RESEARCH INTERESTS

Healthcare AI, Explainable AI, , Deep Learning, Respiratory Signal Analysis, Computational Toxicology,

EDUCATION

Daffodil International University

Bachelor of Science in Software Engineering

CGPA: 3.84/4.00

Undergraduate Thesis: *Interpretable Machine Learning-Based Severity Stratification of Simulated COPD from Respiratory Pressure–Flow Dynamics*

Supervisor: Md. Shohel Arman, Assistant Professor

Ashulia, Dhaka, Bangladesh

July 2022 – August 2026

RESEARCH PROJECTS

Interpretable Machine Learning-Based Severity Stratification of Simulated COPD

2025 – 2026

Undergraduate Thesis Research

- Developed a machine learning-based framework for severity stratification of simulated Chronic Obstructive Pulmonary Disease (COPD) using respiratory pressure–flow dynamics.
- Investigated respiratory features derived from simulated pressure and flow signals for distinguishing different levels of disease severity.
- Evaluated multiple machine learning algorithms for classification and compared their predictive performance.
- Applied explainable AI techniques, including SHAP, to interpret model predictions and identify influential respiratory features.
- Analyzed relationships between respiratory dynamics and model predictions to improve the interpretability of the proposed framework.

SELECTED COURSES

- Data Structures and Algorithms
- Object-Oriented Programming
- Database Management Systems
- Software Engineering
- Computer Networks
- Operating Systems
- Computer Architecture
- Artificial Intelligence
- Machine Learning
- Deep Learning
- Statistics and Probability
- Data Science

TECHNICAL SKILLS

- Programming:** Python, C, C++, Java, R
- Machine Learning:** Scikit-learn, XGBoost, SVM, Logistic Regression, KNN
- Deep Learning:** TensorFlow, PyTorch,
- Data Analysis:** NumPy, Pandas, SciPy
- Data Visualization:** Matplotlib, Seaborn, Plotly
- Development Tools:** Jupyter Notebook, Google Colab, VS Code, Git, GitHub
- Documentation:** LaTeX, Microsoft Office

AWARDS & ACHIEVEMENTS

- Dean's List Award-Spring 2024**

2024

REFERENCES

Md. Shohel Arman

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